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CS 621 DATA ANALYTICS

Numerical Data

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MEAN, MEDIAN, MODE

- Mean:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

- Median:
  - Middle value (avg of the middle two if total length is even)
- Mode: the most frequent element

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VARIANCE AND STANDARD DEVIATION

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}[\sum_{i=1}^{n} x_i^{\,2} - \frac{1}{n}(\sum_{i=1}^{n} x_i)^2]$$

- S^2 is the variance and S is the standard deviation.

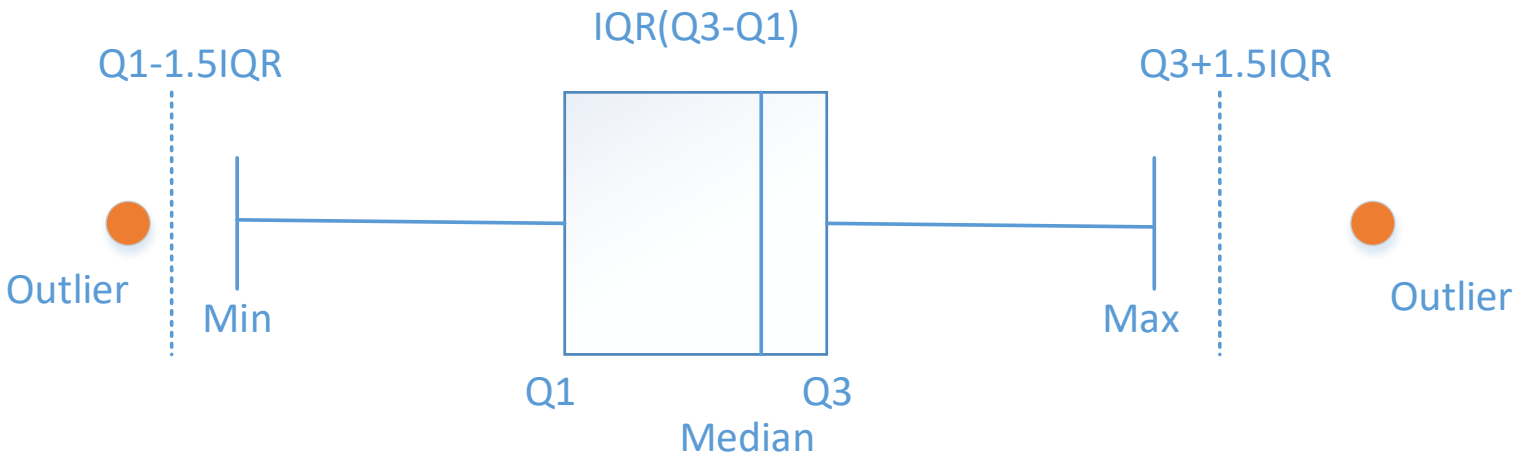
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QUARTILES

- Max
- Min
- Percentiles (25%, 50%, 75%)
- Min, Q1, Q2, Q3, Max
- Q2: median of min and max
- Q1: median of min and Q2
- Q3: median of Q2 and max
- Interquartile range (IQR) = Q3 – Q1
- Outliers: below Q1 – 1.5(IQR) or above Q3+1.5(IQR)

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BOX PLOT



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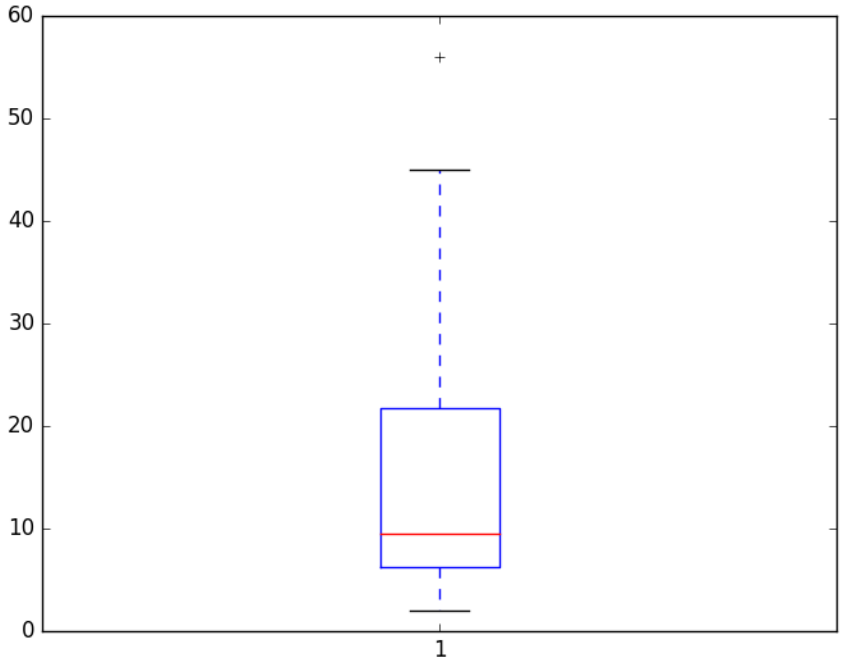
WHAT CAN A BOX PLOT SHOW?

- https://en.wikipedia.org/wiki/Box_plot
  - Q1, Q3, Median
  - One Box
  - Two whiskers: k* IQR, e.g., k = 1.5
  - Outlier(s) if exists
  - Min/Max if no outliers
- Python matplotlib.pyplot boxplot
  - pyplot.boxplot(data, sym='+', vert=1, whis=1.5)

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A BOXPLOT DEMO

- data = [11, 45, 56, 8, 4, 7, 2, 14]
  - pyplot.boxplot(data, sym='+', vert=1, whis=1.5)



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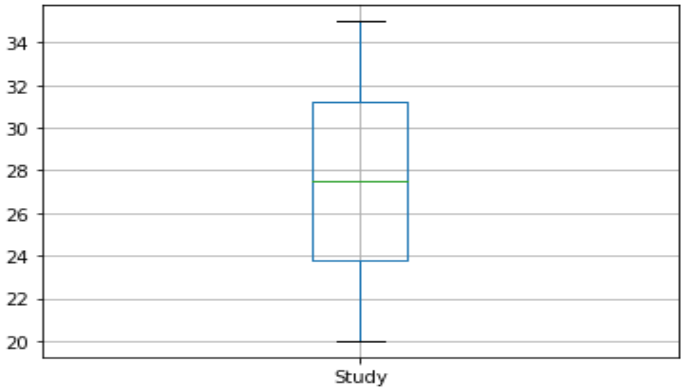
DESCRIPTIVE STATISTICS IN PANDAS SERIES, DATAFRAME

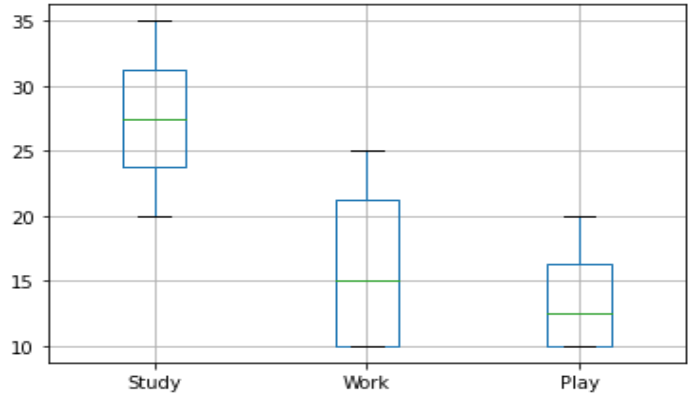
- df = DataFrame (…)
- df[‘column_name’].sum()
- df.sum(axis=1); row by row calculation
- df.mean()
- df.idxmax()
- df.idxmin()
- df.cumsum()
- df.describe()

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BOXPLOT UNDER DATAFRAME

- l = [[20, 10, 10], [25, 10, 15], [30, 20, 10], [35, 25, 20]]
- df = DataFrame(l, index = ['1st', '2nd', '3rd', '4th'], columns= ['Study', 'Work', 'Play'])
- df[‘Study’].describe()
- df.describe()
- df.boxplot(column=“Study”)
- df.boxplot()



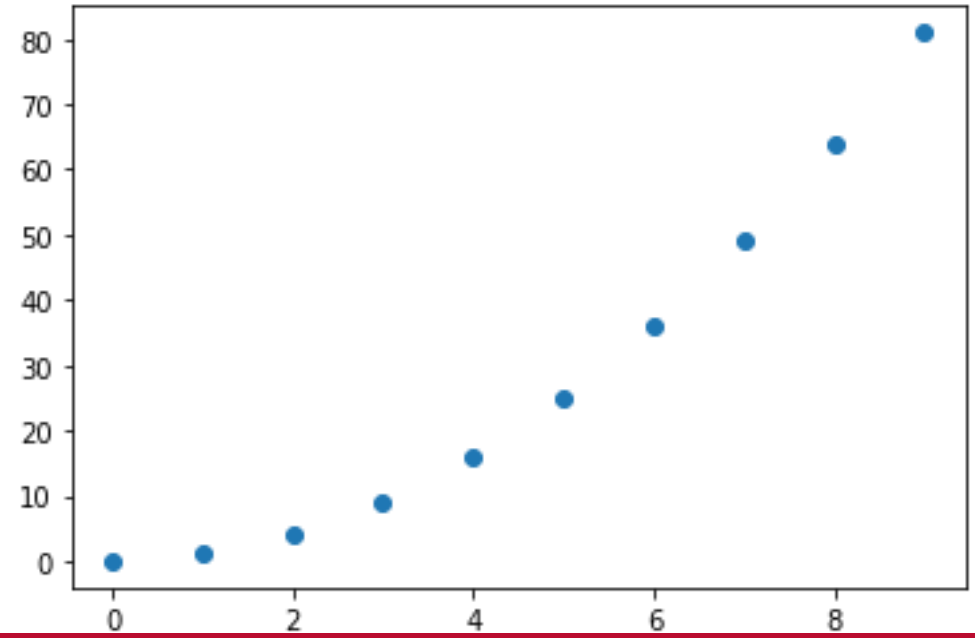


[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]

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SCATTER CHARTS

- X = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
- Y = [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
- plt.scatter(X, Y)
- correlation of two variables
- trend/relationship

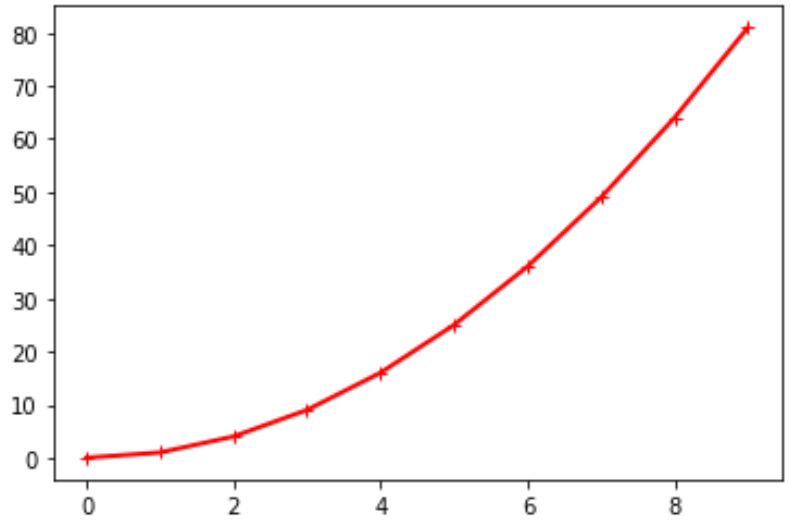


[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]

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LINE CHARTS

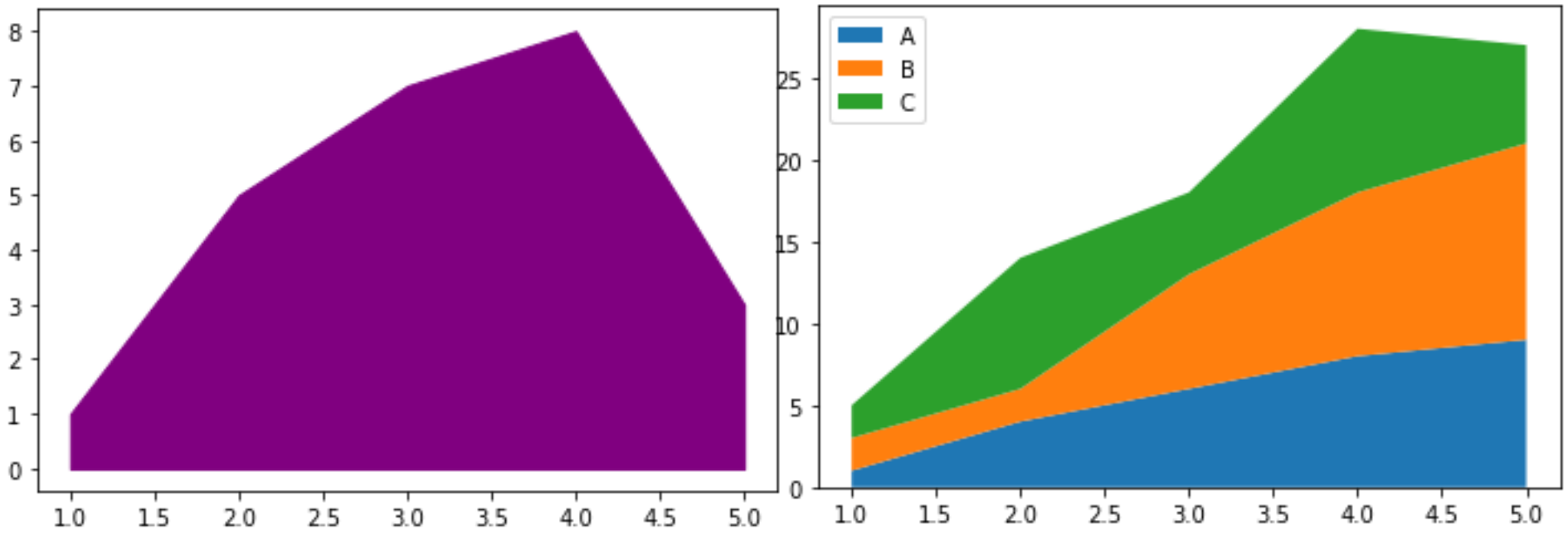
- X = [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
- Y = [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
- plt.plot(X, Y, 'b+', linewidth=2, linestyle="solid", color = "red")
- The x-axis usually represents a sequence of incremental values, e.g., timestamps.



[0, 1, 4, 9, 16, 25, 36, 49, 64, 81]

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AREA AND STACKED AREA CHARTS

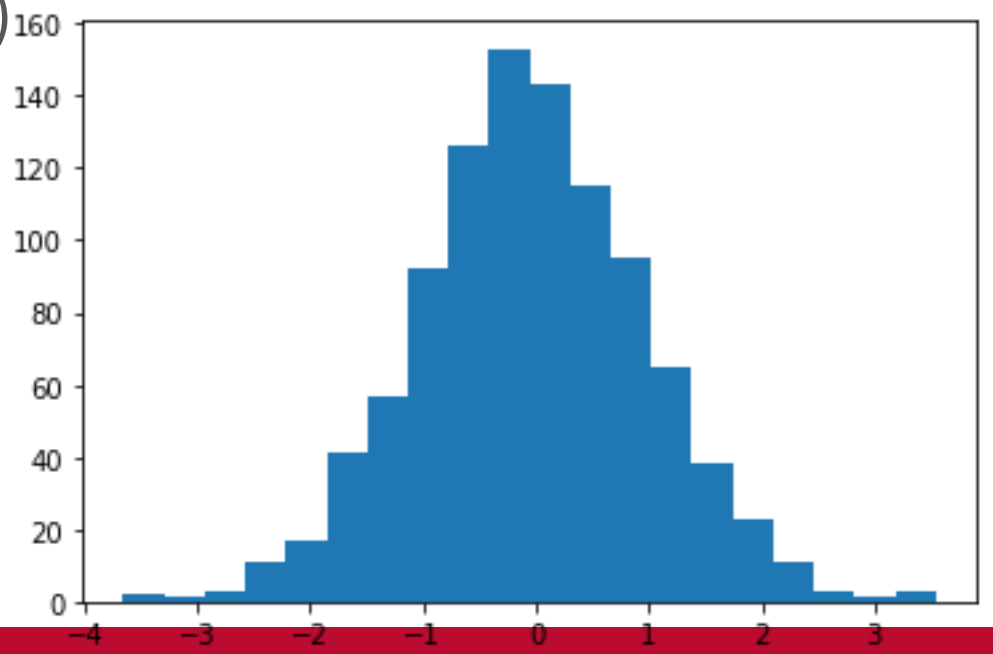


- See code samples for details.

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HISTOGRAMS

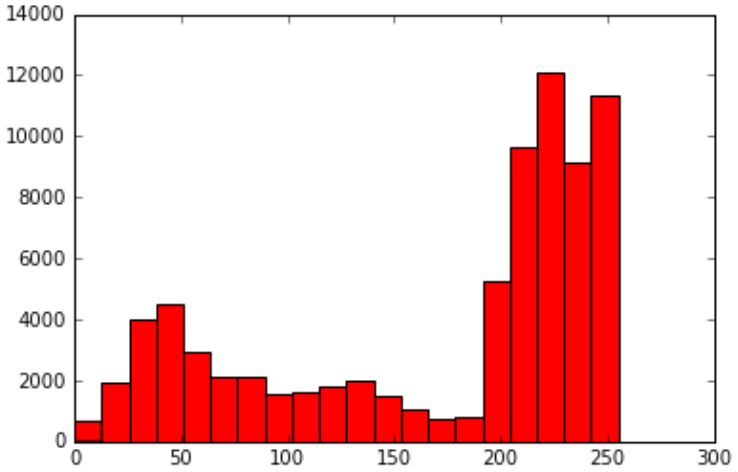
- X axis represents a continuous variable.
- Y axis represents the frequency distribution of values of the variable.
- X = np.random.randn(1000)
- plt.hist(X, bins=20)

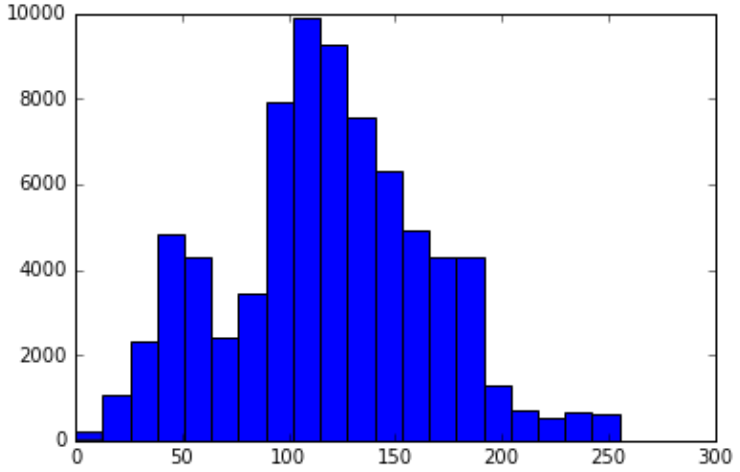


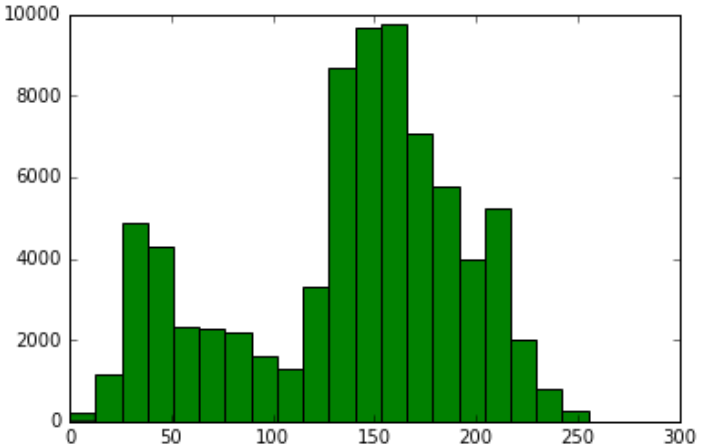
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HISTOGRAMS OF RGB COLORS











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WE FLY